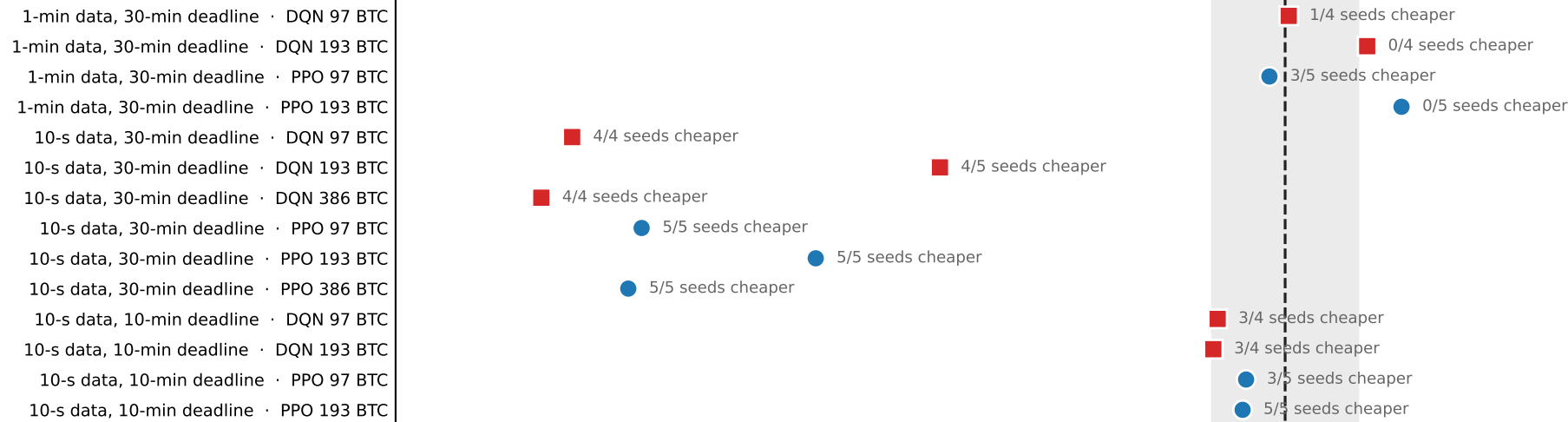


Three environments, three sealed verdicts: no stable improvement on TWAP anywhere -- including where a simple rule proves the edge exists

1. Frozen replay of real order-book data

no stable edge — 4 of 14 arms clear the bar, and a diagnosed-broken learner clears it too



2. Reactive simulator, no injected signal

null — both sealed confirmations fail



3. Reactive simulator + measured signal

null — both regimes fail



-0.6 -0.5 -0.4 -0.3 -0.2 -0.1 0.0 0.1
pooled cost vs TWAP (bps) [left of the line = cheaper than TWAP]

● PPO arm (sealed)

■ DQN arm (sealed)

---- TWAP

±0.05 bps materiality band

■ registered rule, zero fitted parameters

■ trained agents (sealed)

Environment 3: the edge was real
and the agents left all of it

